

SKILL SESSION 1&2

TASK 1:

```
(base) saiabburi@Geyasree-MacBook-Air ~ % gcc --version
Apple clang version 14.0.3 (clang-1403.0.22.14.1)
Target: arm64-apple-darwin22.5.0
Thread model: posix
InstalledDir: /Library/Developer/CommandLineTools/usr/bin
(base) saiabburi@Geyasree-MacBook-Air ~ % git --version
git version 2.39.2 (Apple Git-143)
(base) saiabburi@Geyasree-MacBook-Air ~ % make --version
GNU Make 3.81
Copyright (C) 2006 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.
There is NO warranty; not even for MERCHANTABILITY or FITNESS FOR A
PARTICULAR PURPOSE.

This program built for i386-apple-darwin11.3.0
(base) saiabburi@Geyasree-MacBook-Air ~ % mkdir Skill1
(base) saiabburi@Geyasree-MacBook-Air ~ % cd Skill1
(base) saiabburi@Geyasree-MacBook-Air Skill1 % mkdir src include build
(base) saiabburi@Geyasree-MacBook-Air Skill1 % touch src/main.c
(base) saiabburi@Geyasree-MacBook-Air Skill1 % touch Makefile
```

```
(base) saiabburi@Geyasree-MacBook-Air Skill1 % tree
.
├── Makefile
├── build
├── include
└── src
    └── main.c

4 directories, 2 files
(base) saiabburi@Geyasree-MacBook-Air Skill1 % nano src/main.c
(base) saiabburi@Geyasree-MacBook-Air Skill1 % nano Makefile
(base) saiabburi@Geyasree-MacBook-Air Skill1 % make
gcc -Wall -Wextra -g src/main.c -o build/main
(base) saiabburi@Geyasree-MacBook-Air Skill1 % ./build/main
OSSP Project Started Successfully!
(base) saiabburi@Geyasree-MacBook-Air Skill1 % make run
gcc -Wall -Wextra -g src/main.c -o build/main
./build/main
OSSP Project Started Successfully!
```

UW PICO 5.09

File: src/main.c

```
#include <stdio.h>

int main() {
    printf("OSSP Project Started Successfully!\n");
    return 0;
}
```

UW PICO 5.09

File: Makefile

```
CC = gcc
CFLAGS = -Wall -Wextra -g

TARGET = build/main
SRC = src/main.c

all:
    $(CC) $(CFLAGS) $(SRC) -o $(TARGET)

clean:
    rm -f $(TARGET)

run: all
    ./${TARGET}
```

TASK 2:

```
[(base) saiabburi@Geyasree-MacBook-Air Skill1 % gcc -Wall -Wextra -g src/process_demo.c -o build/process_demo
[(base) saiabburi@Geyasree-MacBook-Air Skill1 % ./build/process_demo
Parent Process Started
Parent PID: 11494
```

Parent waiting for child...

Child Process Created

Child PID: 11495

Parent PID: 11494

Child is executing ls command...

total 8

-rw-r--r-- 1 saiabburi staff 164 Aug 21 23:52 Makefile

drwxr-xr-x 6 saiabburi staff 192 Aug 21 23:59 build

drwxr-xr-x 2 saiabburi staff 64 Aug 21 23:49 include

drwxr-xr-x 4 saiabburi staff 128 Aug 21 23:59 src

Child process completed.

```

Parent process exiting:
(base) saiabburi@Geyasree-MacBook-Air Skill1 % nano src/exec_demo.c
(base) saiabburi@Geyasree-MacBook-Air Skill1 % gcc -Wall -Wextra -g src/exec_demo.c -o build/exec_demo
(base) saiabburi@Geyasree-MacBook-Air Skill1 % ./build/exec_demo
Before exec()
total 8
-rw-r--r--  1 saiabburi  staff   164 Aug 21 23:52 Makefile
drwxr-xr-x  8 saiabburi  staff   256 Aug 22 00:03 build
drwxr-xr-x  2 saiabburi  staff    64 Aug 21 23:49 include
drwxr-xr-x  5 saiabburi  staff   160 Aug 22 00:03 src
(base) saiabburi@Geyasree-MacBook-Air Skill1 % nano src/fork_exec.c
(base) saiabburi@Geyasree-MacBook-Air Skill1 % gcc -Wall -Wextra -g src/fork_exec.c -o build/fork_exec
(base) saiabburi@Geyasree-MacBook-Air Skill1 % ./build/fork_exec
Parent: waiting for child
Child: executing ls
total 8
-rw-r--r--  1 saiabburi  staff   164 Aug 21 23:52 Makefile
drwxr-xr-x 10 saiabburi  staff   320 Aug 22 00:04 build
drwxr-xr-x  2 saiabburi  staff    64 Aug 21 23:49 include
drwxr-xr-x  6 saiabburi  staff   192 Aug 22 00:04 src
Parent: child finished

```

UW PICO 5.09

File: src/process_demo.c

```

#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid;

    printf("Parent Process Started\n");
    printf("Parent PID: %d\n", getpid());

    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        printf("\nChild Process Created\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        printf("Child is executing ls command...\n");

        execlp("ls", "ls", "-l", NULL);

        perror("exec failed");
        exit(1);
    }
    else {
        printf("\nParent waiting for child...\n");

        wait(NULL);

        printf("Child process completed.\n");
        printf("Parent process exiting.\n");
    }

    return 0;
}

```



```
#include <stdio.h>
#include <unistd.h>

int main(void)
{
    printf("Before exec()\n");

    execlp("ls", "ls", "-l", NULL);

    perror("execlp");

    return 1;
}
```

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main(void)
{
    pid_t pid;

    pid = fork();

    if (pid < 0)
    {
        perror("fork");
        return 1;
    }

    if (pid == 0)
    {
        printf("Child: executing ls\n");

        execlp("ls", "ls", "-l", NULL);

        perror("execlp");
        return 1;
    }

    printf("Parent: waiting for child\n");

    wait(NULL);

    printf("Parent: child finished\n");

    return 0;
}
```

```

Parent: child finished
(base) saiabburi@Geyasree-MacBook-Air Skill1 % ps
  PID TTY          TIME CMD
11386 ttys000    0:00.11 -zsh
(base) saiabburi@Geyasree-MacBook-Air Skill1 % ps -ef
  UID    PID  PPID  C STIME  TTY          TIME CMD
  0        1     0  0 Thu07PM ??        3:21.17 /sbin/launchd
  0      299     1  0 Thu07PM ??        1:51.98 /usr/libexec/logd
  0      301     1  0 Thu07PM ??        0:07.09 /usr/libexec/UserEventAgent (System)
  0      303     1  0 Thu07PM ??        0:01.62 /System/Library/PrivateFrameworks/Uninstall.framework/Resources/uninstalld
  0      304     1  0 Thu07PM ??        3:19.97 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks/FSEvents.framework/Vers
  0      305     1  0 Thu07PM ??        0:42.86 /System/Library/PrivateFrameworks/MediaRemote.framework/Support/mediaremoted
  0      308     1  0 Thu07PM ??        0:13.88 /usr/sbin/systemstats --daemon
  0      310     1  0 Thu07PM ??        0:15.38 /usr/libexec/configd
  0      312     1  0 Thu07PM ??        0:32.02 /System/Library/CoreServices/powerd.bundle/powerd
  0      317     1  0 Thu07PM ??        0:00.14 /usr/libexec/remoted
  0      322     1  0 Thu07PM ??        0:01.90 /usr/libexec/watchdogd
  0      326     1  0 Thu07PM ??        0:49.76 /System/Library/Frameworks/CoreServices.framework/Frameworks/Metadata.framework/Support/mds
  0      328     1  0 Thu07PM ??        0:00.96 /usr/libexec/kernelmanagerd
  0      329     1  0 Thu07PM ??        0:44.89 /usr/libexec/diskarbitrationd
  0      333     1  0 Thu07PM ??        0:03.75 /usr/sbin/syslogd
  0      336     1  0 Thu07PM ??        0:07.23 /usr/libexec/thermalmonitord
  0      337     1  0 Thu07PM ??        0:41.59 /usr/libexec/opendirectoryd
  0      338     1  0 Thu07PM ??        0:07.69 /System/Library/PrivateFrameworks/ApplePushService.framework/apsd
  0      339     1  0 Thu07PM ??        0:27.16 /System/Library/CoreServices/launchservicesd
266     340     1  0 Thu07PM ??        0:01.23 /usr/libexec/timed
213     341     1  0 Thu07PM ??        0:00.29 /System/Library/PrivateFrameworks/MobileDevice.framework/Versions/A/Resources/usbmuxd -launchd
  0      342     1  0 Thu07PM ??        0:03.67 /usr/sbin/securityd -i
205     345     1  0 Thu07PM ??        0:55.05 /usr/libexec/locationd
  0      347     1  0 Thu07PM ??        0:00.01 autofs
  0      348     1  0 Thu07PM ??        0:29.61 /usr/libexec/das
  0      349     1  0 Thu07PM ??        0:00.10 /System/Library/PrivateFrameworks/Heimdal.framework/Helpers/kdc
241     351     1  0 Thu07PM ??        0:15.08 /usr/sbin/distnoted daemon
  0      355     1  0 Thu07PM ??        0:00.21 /System/Library/CoreServices/login
  0      356     1  0 Thu07PM ??        0:03.79 /System/Library/PrivateFrameworks/GenerationalStorage.framework/Versions/A/Support/revisi

```

TASK 3:

```

(base) saiabburi@Geyasree-MacBook-Air Skill1 % nano src/myshell.c
(base) saiabburi@Geyasree-MacBook-Air Skill1 % gcc -Wall -Wextra -g src/myshell.c -o build/myshell
(base) saiabburi@Geyasree-MacBook-Air Skill1 % ./build/myshell
Mini Shell Started
Type 'exit' to quit.

myshell> Hello
You entered: Hello
myshell> Linux
You entered: Linux
myshell> exit
Exiting shell...

```

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX_INPUT 100

int main() {
    char input[MAX_INPUT];

    printf("Mini Shell Started\n");
    printf("Type 'exit' to quit.\n\n");

    while (1) {
        printf("myshell> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL) {
            break;
        }

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0) {
            printf("Exiting shell...\n");
            break;
        }

        if (strlen(input) == 0) {
            continue;
        }

        printf("You entered: %s\n", input);
    }

    return 0;
}
```

TASK 4:

```
(base) saiabburi@Geyasree-MacBook-Air Skill1 % nano src/keyboard_shell.c
(base) saiabburi@Geyasree-MacBook-Air Skill1 % gcc -Wall -Wextra -g src/keyboard_shell.c -o build/keyboard_shell
(base) saiabburi@Geyasree-MacBook-Air Skill1 % ./build/keyboard_shell
Keyboard Input Shell
Type commands and press Enter.
Type 'exit' to quit.

[myshell> HELLO
Command entered: HELLO
[myshell> hello world
Command entered: hello world
[myshell> exit
Exiting shell...
```


(base) saiabburi@Geyasree-MacBook-Air Skill11 % tree

```
.
├── Makefile
├── build
│   ├── exec_demo
│   │   ├── exec_demo.dSYM
│   │   │   └── Contents
│   │   │       ├── Info.plist
│   │   │       ├── Resources
│   │   │       │   └── DWARF
│   │   │           └── exec_demo
│   ├── fork_exec
│   │   ├── fork_exec.dSYM
│   │   │   └── Contents
│   │   │       ├── Info.plist
│   │   │       ├── Resources
│   │   │       │   └── DWARF
│   │   │           └── fork_exec
│   ├── keyboard_shell
│   │   ├── keyboard_shell.dSYM
│   │   │   └── Contents
│   │   │       ├── Info.plist
│   │   │       ├── Resources
│   │   │       │   └── DWARF
│   │   │           └── keyboard_shell
│   ├── main
│   │   ├── main.dSYM
│   │   │   └── Contents
│   │   │       ├── Info.plist
│   │   │       ├── Resources
│   │   │       │   └── DWARF
│   │   │           └── main
│   ├── myshell
│   │   ├── myshell.dSYM
│   │   │   └── Contents
│   │   │       ├── Info.plist
│   │   │       ├── Resources
│   │   │       │   └── DWARF
│   │   │           └── myshell
│   ├── process_demo
│   │   ├── process_demo.dSYM
│   │   │   └── Contents
│   │   │       ├── Info.plist
│   │   │       ├── Resources
│   │   │       │   └── DWARF
│   │   │           └── process_demo
├── include
└── src
    ├── exec_demo.c
    ├── fork_exec.c
    ├── keyboard_shell.c
    ├── main.c
    ├── myshell.c
    └── process_demo.c
```

28 directories, 25 files

```
#include <stdio.h>
#include <unistd.h>
#include <termios.h>
#include <string.h>

#define MAX_INPUT 100

void enable_raw_mode(struct termios *original) {
    struct termios raw;

    tcgetattr(STDIN_FILENO, original);
    raw = *original;

    raw.c_lflag &= ~(ICANON | ECHO);

    tcsetattr(STDIN_FILENO, TCSAFLUSH, &raw);
}

void disable_raw_mode(struct termios *original) {
    tcsetattr(STDIN_FILENO, TCSAFLUSH, original);
}

int main() {
    struct termios original;
    char buffer[MAX_INPUT];
    int position = 0;
    char ch;

    printf("Keyboard Input Shell\n");
    printf("Type commands and press Enter.\n");
    printf("Type 'exit' to quit.\n\n");

    enable_raw_mode(&original);

    while (1) {
        printf("myshell> ");
        fflush(stdout);

        position = 0;

        while (1) {
            if (read(STDIN_FILENO, &ch, 1) <= 0) {
                break;
            }
        }
    }
}
```



```

        if (ch == 127 || ch == '\b') {
            if (position > 0) {
                position--;

                printf("\b \b");
                fflush(stdout);
            }

            continue;
        }

        if (position < MAX_INPUT - 1) {
            buffer[position++] = ch;
            printf("%c", ch);
            fflush(stdout);
        }

        if (strcmp(buffer, "exit") == 0) {
            printf("Exiting shell...\n");
            break;
        }

        if (position > 0) {
            printf("Command entered: %s\n", buffer);
        }
    }
}

```

```

        if (position < MAX_INPUT - 1) {
            buffer[position++] = ch;
            printf("%c", ch);
            fflush(stdout);
        }
    }

    if (strcmp(buffer, "exit") == 0) {
        printf("Exiting shell...\n");
        break;
    }

    if (position > 0) {
        printf("Command entered: %s\n", buffer);
    }
}

disable_raw_mode(&original);

return 0;
}

```

SIMPLE COMMANDS

```
Last login: Mon Aug 10 17:18:18 on ttys000
(base) saiabburi@Geyasree-MacBook-Air ~ % hostname
Geyasree-MacBook-Air.local
(base) saiabburi@Geyasree-MacBook-Air ~ % id
uid=501(saiabburi) gid=20(staff) groups=20(staff),12(everyone),61(localaccounts),79(_appserverusr),80(admin),81(_appserveradm),98(_lpadmin),33(_appstore),100(_lpoperator),204(_developer),250(_analyticsusers),395(com.apple.access_ftp),398(com.apple.access_screensharing),399(com.apple.access_ssh),400(com.apple.access_remote_ae),701(com.apple.sharepoint.group.1)
(base) saiabburi@Geyasree-MacBook-Air ~ % tty
/dev/ttys000
(base) saiabburi@Geyasree-MacBook-Air ~ % uptime
19:20 up 44 mins, 2 users, load averages: 1.88 1.86 1.99
(base) saiabburi@Geyasree-MacBook-Air ~ % users
saiabburi
(base) saiabburi@Geyasree-MacBook-Air ~ % who
saiabburi      console    Aug 10 18:36
saiabburi      ttys000    Aug 10 19:19
(base) saiabburi@Geyasree-MacBook-Air ~ % w
19:20 up 44 mins, 2 users, load averages: 1.90 1.87 1.99
USER      TTY      FROM      LOGIN@   IDLE WHAT
saiabburi console -          18:36    43 -
saiabburi s000     -          19:19    - w
```

```
(base) saiabburi@Geyasree-MacBook-Air ~ % groups
staff everyone localaccounts _appserverusr admin _appserveradm _lpadmin _appstore _lpoperator _developer _analyticsusers com.apple.access_ftp com.apple.access_screensharing com.apple.access_ssh com.apple.access_remote_ae com.apple.sharepoint.
group.1
(base) saiabburi@Geyasree-MacBook-Air ~ % env
__CFBundleIdentifier=com.apple.Terminal
TMPDIR=/var/folders/kt/010117zs6wv1hzwk1chttmm000gn/T/
XPC_FLAGS=0x0
TERM=xterm-256color
SSH_AUTH_SOCK=/private/tmp/com.apple.launchd.xa600vier3/Listeners
XPC_SERVICE_NAME=0
TERM_PROGRAM=Apple_Terminal
TERM_PROGRAM_VERSION=447
TERM_SESSION_ID=4F8A395E-09DE-4F16-B837-0B6E883D5244
SHELL=/bin/zsh
HOME=/Users/saiabburi
LOGNAME=saiabburi
USER=saiabburi
PATH=/opt/anaconda3/bin:/opt/anaconda3/condabin:/usr/local/bin:/System/Cryptexes/App/usr/bin:/usr/bin:/bin:/sbin:/sbin:/Applications/VMware Fusion.app/Contents/Public:/opt/homebrew/bin:/var/run/com.apple.security.cryptexd/codex.system/boo
tstrap:/usr/local/bin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/bin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/appleinternal/bin
SHLVL=1
PWD=/Users/saiabburi
OLDPWD=/Users/saiabburi
CONDA_EXE=/opt/anaconda3/bin/conda
_CE_M=
_CE_CONDA=
CONDA_PYTHON_EXE=/opt/anaconda3/bin/python
CONDA_SHLVL=1
CONDA_PREFIX=/opt/anaconda3
CONDA_DEFAULT_ENV=base
CONDA_PROMPT_MODIFIER=(base)
GSETTINGS_SCHEMA_DIR_CONDA_BACKUP=
GSETTINGS_SCHEMA_DIR=/opt/anaconda3/share/glib-2.0/schemas
LC_CTYPE=UTF-8
_=usr/bin/env
(base) saiabburi@Geyasree-MacBook-Air ~ % df
Filesystem      512-blocks    Used Available Capacity  iused   ifree %iused   Mounted on
/dev/disk3s1s1  478724992    23609408    10446816    70%   355382  52234080    1%   /
devfs              404          404          0    100%    702      0    100%   /dev
/dev/disk3s6      478724992      48    10446816     1%      0  52234080     0%   /System/Volumes/VM
/dev/disk3s4      478724992   19280664    10446816    65%   1202  52234080     0%   /System/Volumes/Preboot
/dev/disk3s2      478724992   1471176    10446816    13%    299  52234080     0%   /System/Volumes/Update
/dev/disk1s2      1024000     12328    982944     2%      1  4914720     0%   /System/Volumes/xarts
/dev/disk1s1      1024000     12800    982944     2%     41  4914720     0%   /System/Volumes/ISCPreboot
/dev/disk1s3      1024000      6208    982944     1%     63  4914720     0%   /System/Volumes/Hardware
/dev/disk3s1      478724992  420484800    10446816    98%  1318436  52234080     2%   /System/Volumes/Data
map auto_home          0          0          0    100%      0      0    100%   /System/Volumes/Data/home
/dev/disk3s3      478724992    23609408    10446816    70%   356884  52234080     1%   /System/Volumes/Update/mt1
```

```
(base) saiabburi@Geyasree-MacBook-Air ~ % free
zsh: command not found: free
(base) saiabburi@Geyasree-MacBook-Air ~ % history
[ 143  echo "Hello World"
[ 144  sudo apt install bc
[ 145  bc
[ 146  factor 1234
[ 147  sleep 5
[ 148  hostname
[ 149  id
[ 150  tty
[ 151  uptime
[ 152  users
[ 153  who
[ 154  w
[ 155  groups
[ 156  env
[ 157  df
[ 158  free
(base) saiabburi@Geyasree-MacBook-Air ~ % uname -r
22.5.0
(base) saiabburi@Geyasree-MacBook-Air ~ % iostat
[          disk0          cpu          load average
[      KB/t  tps  MB/s  us sy id    1m   5m   15m
[    12.87  347   4.36    6  3  91   1.96  1.94  2.00
(base) saiabburi@Geyasree-MacBook-Air ~ % echo "Hello World"
Hello World
(base) saiabburi@Geyasree-MacBook-Air ~ % bc
[>>> 12233344444
12233344444
[>>> quit
[ (base) saiabburi@Geyasree-MacBook-Air ~ % find
usage: find [-H | -L | -P] [-EXdsx] [-f path] path ... [expression]
[ find [-H | -L | -P] [-EXdsx] -f path [path ...] [expression]
[ (base) saiabburi@Geyasree-MacBook-Air ~ % █
```